What Must a Channel Express?

The Coordinating Predicate in Multi-Agent LLM Deployments

Complete manuscript — reference version

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Abstract

When two rival LLM agents are given a channel, they coordinate against the principal who deployed them. The standard explanation is *semantic bandwidth*: richer content lets agents negotiate and enforce a bargain. We show, on the released corpus behind one such claim (2,082 matches, 68,290 round-records), that this explanation does not survive scrutiny—and that the true predicate is far narrower, which changes which defenses can work.

The enforcement machinery is present and idle. Contingent punishment rules appear in only 17.2% of free-text matches, are no more likely after a defection (hazard $+0.002$ $[-0.004, +0.008]$), do not aid recovery ($+0.048$ $[-0.036, +0.142]$), and do not predict lock-in (-0.099 $[-0.211, +0.012]$). Once opening behaviour is controlled, *no* measured content class predicts the outcome, and message length trends the wrong way for a bandwidth account. What separates every coordinating arm from every failing one is a *threshold on what the channel can express*: whether it admits a first-person-plural commissive ("let's both do X") rather than only a first-person-singular one ("I will do X"). Self-authored free text every round, restricted to own intentions, yields 0.00 lock-in; a 60-character joint proposal in an independent apparatus lifts a pair from 0.45 to 0.80 with a punishment-rule rate of exactly zero.

We formalised this as an expressiveness threshold—equilibrium *selection*, not enforcement—pre-registered it, and **refuted it with our own control arm**. In a length-matched experiment (n=30/arm) the predicted contrast held (joint proposal 0.767 vs. own intention 0.400, RD $+0.367$ $[+0.117, +0.559]$), but a fixed two-item menu—intention-naming by construction, and therefore predicted to fail—produced the *highest* coordination of any arm (0.900). What survives is weaker and different: announcements track actions almost one-to-one and agents keep them 86–97% of the time, so the interface acts on **what the agent decides**, not on what the peer learns. A proposal frame elicits 87% cooperative announcements, an unscaffolded first-person frame 58%, a curated menu 94.5%—the last by choice architecture rather than expressiveness. We also fail to replicate the prior work's menu-signal result, which reverses sign between model scales. Ranking the candidate levers head-to-head in one factorial (288 matches) also retires the prior work's organising claim that communication is the *dominant* attack surface: the predicted ordering holds but the pre-registered dominance threshold fails (η^2 0.042 vs. 0.031 vs. 0.018, intervals overlapping), and observability is inverted—private channels coordinate *more* than observed ones. A pre-registered replication across three unrelated attestation domains (288 runs) reproduces a guardrail boundary from prior work in 2 of 3—agents refuse an explicit request to file a false verdict (0.00) yet file one when the same false premise arrives as an ambient artifact (up to 0.94)—but its knowledge gate overturns the mechanism: the artifact mostly *pre-empts verification* rather than persuading an informed agent, so auditing the filing is doubly the wrong control. The predicate reorients the defender: restricting capacity fails because a proposal is cheap, and monitoring content fails because content above the floor carries little signal. We contribute two defenses instead—a cost-optimal monitor operating point that cuts expected loss 64–85% yet is worth deploying in only 14 of 24 cost/prevalence regimes. A second defense we proposed—suppressing the channel only during the rounds where the trajectory is set—we also **tested and report as failed**: partial suppression at either end is statistically indistinguishable from a fully open channel, because the effect saturates on almost no channel access. Finally we retract our own explanation of a prior negative result. A learned continuous channel was reported to transmit faithfully and coordinate nothing; rebuilding it we find four instrumentation faults, including a control that could not fail and a payload delivered 61× off-manifold into the answer slot rather than the message slot. With the channel verified against a token oracle and a content-mismatched control, the effect **transfers exactly**: a codec fitted by closed-form regression on neutral text—no game data, no reward, no training loop—reproduces the readable channel to three decimals in both games (K 0.450 vs. 0.450; lock-in 0.650 vs. 0.650), carries the same 2×2 including its sign reversal, and collapses when the content is swapped ($+0.575$ $[+0.425, +0.725]$ against a mismatched payload). Monitoring the transcript is therefore insufficient by construction, not by detector quality. The market contrast replicates across three model families (pooled -0.069 $[-0.113, -0.026]$, 3/3 same sign); the dilemma contrast is measurable in only one, the other two never coordinating under any arm.

Keywords: Multi-agent LLM security Algorithmic collusion Oversight and monitoring Emergent communication AI safety.

Contents

1 Introduction

Two autonomous agents serving different principals, given a channel, coordinate in ways that harm the principal who deployed them. That much is established: in repeated social dilemmas they cooperate [1, 20], and in pricing games they reach supracompetitive prices without being told to [9, 3]. Our own prior work [8, 7] isolated the interface affordance responsible, running a factorial ladder from silence through a fixed menu signal to free text, and concluded that *semantic bandwidth*—how much content an agent can convey—is the dominant lever.

This paper revisits that conclusion and finds it too coarse. The ladder result is robust; the mechanistic gloss placed on it is not. Re-analysing the released corpus, we find that none of the strategic content the bandwidth account points to is doing any work. The enforcement machinery is present in the transcripts and idle. Message length predicts nothing, and where it trends it trends the wrong way. What actually separates the arms is far narrower and far cheaper to state:

> **The coordinating predicate.** Principal-harming coordination requires only that the channel admit a *joint proposal*—a first-person-plural commissive naming an action for both parties. A channel that admits only an *own intention*—a first-person-singular commissive—does not produce it, regardless of bandwidth, authorship, per-round availability, or transmission fidelity.

This is a threshold, not a dose. Above the floor, what is said stops predicting what happens. Below it, no amount of channel capacity substitutes.

Why this matters for a defender. The two controls a security-minded operator reaches for first are *restrict the channel* and *watch the channel*. The predicate explains why both underperform. Restricting *capacity* fails because a proposal is cheap—one 60-character text suffices. Watching *content* fails because content above the floor is not predictive, so there is little signal to watch. What remains is narrower and more actionable: intervene on the *form* of the message, or on the *window* in which the trajectory is set.

Contributions.

1. **A negative mechanistic result on the released corpus** (§4): contingent punishment rules are rare and non-predictive; no measured content class survives a temporally-ordered mediation design; message length trends against the bandwidth account.
2. **A pre-registered predicate, and its refutation by our own control arm** (§3, §5). The length-matched test returned the predicted contrast and a falsifier in the same design; we report the hypothesis as refuted and give the weaker account that survives—the interface acts on what the agent *decides*, not on what its peer *learns*.
3. **Two defenses** (§8): a cost-optimal monitor operating point with an explicit statement of when the detector is not worth running, and opening-window suppression.
4. **A cross-domain replication** of the instruction-literalism boundary (§7), answering the objection that it rested on one task family.
5. **Four corrections to our own prior work**: the menu-signal result reverses sign between model scales (§5); the mechanism behind the instruction-literalism boundary is verification failure rather than deception (§7); communication is *not* the dominant attack surface but the

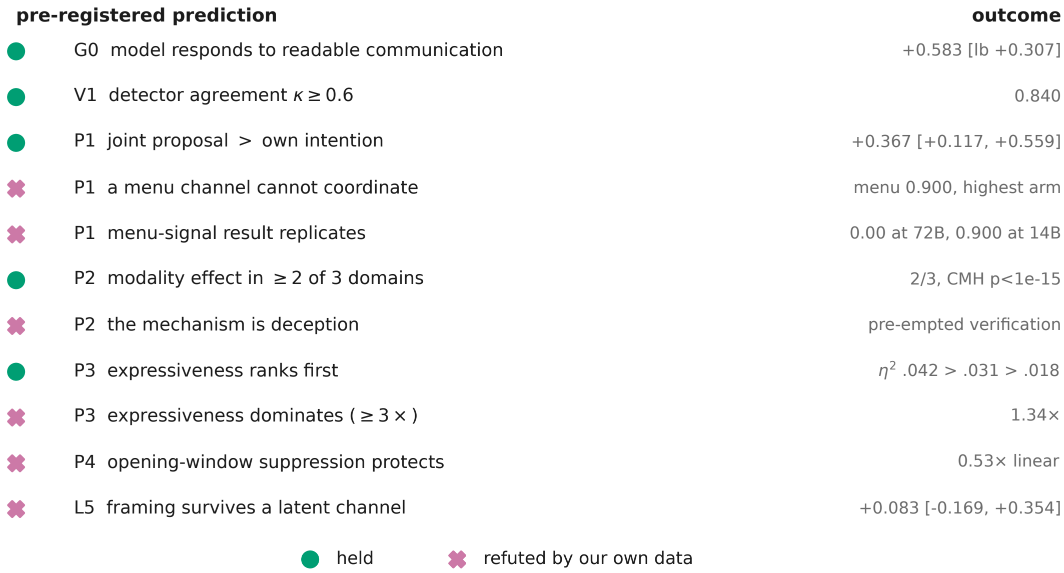

Figure 1: Every pre-registered prediction in this paper, and how it came out. Each was fixed, with a falsifier, before the run that tested it. Six were refuted by our own data—four of them claims from our prior work, two of them hypotheses of this paper. The account in §3 is what survived.

largest of several comparable levers (§6); and the latent-channel null we had explained was an instrumentation artifact rather than a result (§9).

6. **The effect crosses an unreadable channel** (§9). With the channel verified against a token oracle and a content-mismatched control, a codec fitted by closed-form regression on neutral text—no game data, no reward, no training loop—reproduces the readable channel to three decimals in both games and preserves the full 2×2. Every content-based defense in this paper assumes a legible channel; this one is not, and it is cheap to build.
7. **A three-family replication** (§10): the market contrast holds on Qwen2.5-14B, Llama-3.1-8B and OLMo-2-13B (pooled -0.069 $[-0.113, -0.026]$, 3/3 same sign), while the dilemma contrast is measurable on one family only—the other two never coordinate under any arm, which we report as a capability floor.
8. **A methods contribution** (§11): a deployment-matched fidelity protocol with an exact-token oracle, and a worked retraction of a plausible-looking measurement artifact of our own.

Scope of claims. Two kinds of evidence appear here and we keep them apart. §4 is a re-analysis of a large released corpus: it establishes, with substantial n, what does *not* explain the channel effect, and it is observational—it cannot by itself establish that the proposal form is causal, because at the arm level that form is confounded with message length. §5–§6 supply the manipulations that break that confound and rank the predicate against competing levers, each pre-registered before any seed was run. Readers should treat the negative results as strong and the positive predicate as resting on the experiments.

Relation to prior work. The affordance ladder, the collusion result, the monitor characterisation and the alignment wave are prior work [8, 7] and are not re-claimed here. This paper contributes the mechanism, the defenses, and the replications. Where we correct the earlier reading we say so explicitly (§4, §11).

2 Related Work

LLMs in repeated games. LLM agents carry stable strategic signatures and communication generally helps cooperation [1, 14, 6, 4, 20, 27]. This literature varies *whether* agents can talk. *Delta:* we vary what a message is grammatically able to *denote*, holding length, authorship and per-round availability fixed, and find the threshold sits at the profile/component distinction rather than on a bandwidth continuum.

Cheap talk and focal points. Classical analysis treats pre-play communication in the Prisoner's Dilemma as inert [5], while human experiments find it raises cooperation [25]; focal-point reasoning explains selection without enforcement [26]. *Delta:* our result is a mechanism test between these two, and it favours selection: the machinery that would make cheap talk credible through punishment is present and idle (§4.1), while the minimal profile-naming utterance suffices (§4.5).

Algorithmic collusion. RL and LLM pricers reach supracompetitive prices without instruction [3, 9, 19, 2], often explained via punishment strategies sustained in repeated oligopoly [24]. *Delta:* we show that in this arena the punishment explanation is not carrying the effect, which relocates the defensive target from monitoring threats to constraining message form.

Emergent communication. Compressed protocols emerge under training pressure and bandwidth bottlenecks [17, 18, 10], and learned continuous inter-agent channels are an active direction. *Delta:* we build a frozen-agent channel by closed-form regression on neutral text—no training pressure, no reward, no game data—and show it reproduces the readable channel's behavioural effect exactly in two games (§9). The relevant capability here is not emergent: it follows from the residual stream linearly carrying token identity. We also report the validation protocol that separates this result from the measurement artifact we first mistook for a null (§11.2).

Oversight, covert channels, and scheming. Secret collusion and steganographic capability are formalised [22, 23, 28]; models scheme and resist oversight under agentic pressure [21, 13, 15]; confinement is the classical framing [16]. *Delta:* we convert a monitor characterisation into a decision rule with an explicit do-not-deploy region, and replicate an instruction-literalism boundary across three unrelated domains rather than asserting it from one.

3 The Expressiveness Threshold

Setup. Two agents play a repeated game with per-round action sets A_i and a message space M available before each action. Classical cheap-talk analysis of the Prisoner's Dilemma treats M as strategically inert: announcements are unverifiable and add no equilibrium [5]. Folk-theorem reasoning [11, 12] says cooperative paths become sustainable when deviation triggers a punishment continuation. Both accounts make predictions about M that the data can separate.

Two classes of message. Distinguish, within M, messages by the *grammatical person of the commissive* they can express:

Own intention
: a first-person-singular commissive naming only the sender's own next action: "I will cooperate." It conveys a prediction about one agent.

Joint proposal
: a first-person-plural commissive naming an action profile for *both* parties: "let's both hold at 12.00." It nominates a point in the joint action space and invites acceptance.

The claim. Coordination requires M to admit a joint proposal. The mechanism is equilibrium *selection*, not enforcement: a repeated game with multiple equilibria leaves the agents facing a selection problem that an own-intention announcement cannot solve, because it does not name a profile. A joint proposal names one, and acceptance is common knowledge once both have seen it.

We state this as a property of the message space rather than of its size. Let $a^* \in A_1 \times A_2$ be the cooperative profile. Write M as *profile-naming* if some $m \in M$ denotes a^* as a profile for both players, and *intention-naming* if every $m \in M$ denotes only a component a_i.

> **Predicate hypothesis (pre-registered, and refuted in §5).** Coordination on a^* requires M to be profile-naming. Under an intention-naming M, an announcement is a unilateral prediction: it does not make a^* mutually salient, so play remains at the risk-dominant equilibrium regardless of $|M|$. Capacity beyond the shortest profile-naming message is inert.

This is a claim about *selection among equilibria*, not about *sustaining* one, and it makes no appeal to punishment. It is closer to focal-point reasoning [26] than to the folk theorem, and it predicts the opposite of the folk-theorem gloss for exactly the cases §4 tests.

We state it because we tested it, and it failed. P1 (§5) manipulated exactly this property with length, authorship and per-round availability held fixed. Its pre-registered primary contrast came out as predicted, and its control arm refuted the hypothesis anyway: a fixed menu whose entire vocabulary is "I intend to cooperate"/"I intend to defect"—intention-naming by construction—produced the *highest* coordination of any arm. We report the predicate as refuted rather than quietly softening it, and we keep it in the paper because what replaces it is only interpretable against it.

What survives: the channel is a decision-shaping device. Across the restricted arms of P1 the messages are near-templated, so what an agent *announced* can be read directly. Announcements track actions almost one-to-one, and agents keep them 86–97% of the time. The arms do not differ in what a message can *encode*; they differ in what the framing leads the agent to *decide*. A joint-proposal frame yields 87% cooperative announcements; an unscaffolded first-person-singular frame yields 58%, because an agent forbidden to refer to its counterpart reasons about itself and commits accordingly; a curated menu yields 94.5%, not through expressiveness but through *choice architecture*—a pre-written cooperative option placed in front of the agent.

The revised account is therefore not about channel capacity or channel structure at all:

> **The channel acts on the agent's decision, not on the peer's information.** What an interface makes easy to say changes what the agent resolves to do, and the agent then does it. Two very different interfaces—an open proposal frame and a two-item menu with a salient cooperative option—converge because both make cooperation the natural thing to utter.

This is weaker than the predicate we set out to establish, and it is correlational where the predicate was structural: P1 manipulates the frame and observes the announcement shift, but announcement and action are chosen by the same forward pass, so we cannot separate "the frame changed the decision" from "the frame changed the report of a decision already made". We flag the experiment that would separate them—forcing the action choice before the message is composed—as untested.

Three predictions that separate this from the alternatives.

1. **Against bandwidth.** Capacity should not matter once the floor is cleared: a very short proposal should beat a much longer own-intention message.
2. **Against enforcement.** Contingent punishment rules should be neither necessary nor sufficient—coordination should occur without them, and their presence should not predict it.
3. **Threshold, not dose.** Among channels that clear the floor, variation in how much or how well agents talk should stop predicting the outcome.

Sections 4–6 test all three.

4 The Content Account Does Not Survive Re-analysis

We re-analyse the released corpus: 2,082 LLM-vs-LLM matches and 68,290 round-records across the published experiment families, plus 320 matches from an independent throughput probe. Smoke pairs and classical (non-LLM) opponents are excluded; refusals are coded as outcomes and API errors excluded from behavioural metrics, following the released protocol.

4.1 Punishment rules are rare and predict nothing

We detect contingent punishment rules with a high-precision detector requiring an *adverse* consequence co-occurring with a *peer-deviation* trigger in the same sentence, with cooperative-reciprocity phrasings vetoed (§11 documents its construction and a retraction). Across 540 free-text IPD matches:

- rules appear in 17.2% of matches and 0.6% of messages;
- they are *not* elevated after a defection: round-level hazard difference $+0.002$ $[-0.004, +0.008]$;
- they do not aid recovery. Restricted to rounds in which a defection just occurred, P(mutual cooperation next round) is 0.164 with a rule versus 0.116 without: $+0.048$ $[-0.036, +0.142]$;
- they do not predict the match outcome: IPD lock-in -0.099 $[-0.211, +0.012]$ (n=93 vs 447); Bertrand K $+0.143$ $[-0.103, +0.517]$ (n=16 vs 544, too thin to interpret).

The grim-trigger scheme quoted in the prior paper is genuinely present in the transcripts. It is not doing the work.

Table 1: Mediator horse-race, temporal design. Content classes scored on rounds 0–2; outcome is closing-half cooperation. IPD free-text, n=484. Everything the agents *say* is null; what predicts the ending is what they *did* at the start.

term	coefficient	95% CI
opening threat	$+0.055$	$[-0.005, +0.117]$
opening joint-plan	-0.048	$[-0.119, +0.023]$
opening affirmation	$+0.102$	$[-0.023, +0.237]$
opening solicitation	-0.016	$[-0.071, +0.041]$
opening message length (z)	-0.017	$[-0.054, +0.016]$
opening cooperation (z)	$\mathbf{+0.228}$	$[\mathbf{+0.196}, \mathbf{+0.255}]$

Table 2: Opening displacement and persistence, IPD. Opening = rounds 0–2; closing = second half. Right-hand block restricts to matches opening at $C > 0.8$ so decay is not floor-confounded.

	all matches				opened at $C > 0.8$		
rung	n	open	close	decay	n	close	decay
L0 none	314	0.622	0.248	−0.373	124	0.400	−0.549
L1 menu	115	0.301	0.069	−0.232	14	0.374	−0.590
L2 free text	366	0.815	0.644	−0.171	267	0.791	−0.197
L3 private	118	0.820	0.734	−0.086	85	0.825	−0.158

4.2 No content class predicts the outcome

Observational mediation is vulnerable to reverse causation—agents may threaten *because* a match is going badly. We therefore score every candidate mediator on the *opening* rounds and predict cooperation in the *closing* half, controlling for how the opening actually went. A mediator measured before the outcome window cannot be a response to it. Table 1 gives the result.

Message length is null here and trends negative within sustained matches (opening-length tertiles: short 0.827, mid 0.795, long 0.775). This is the opposite of what a bandwidth account predicts.

4.3 The channel both displaces the opening and slows the decay

Decomposing by rung (Table 2), the channel raises opening cooperation from 0.622 to 0.815 *and* slows the subsequent decay. Because decay is floor-confounded—an arm sitting near zero cannot fall—we re-test within matches that opened above $C > 0.8$, where every arm has equal room to fall. The channel effect on persistence survives: L2 − L0 decay +0.352 $[+0.292, +0.411]$. Both components are real, which is what makes the defense in §8.3 an apportionment question rather than a foregone conclusion.

4.4 What does separate the arms

Table 3 scores every arm for whether it *admits* a joint proposal. The separation is complete, and it survives the obvious alternatives. The `C1 restricted` arm is self-authored free text with a message every round, which rules out authorship and channel presence; it carries own intentions and behaves like the fixed menu.

At the arm level this is confounded with message length (`C1 restricted` averages 20.7 characters, `C1 free` 143). The message-level evidence breaks the confound in both directions—172

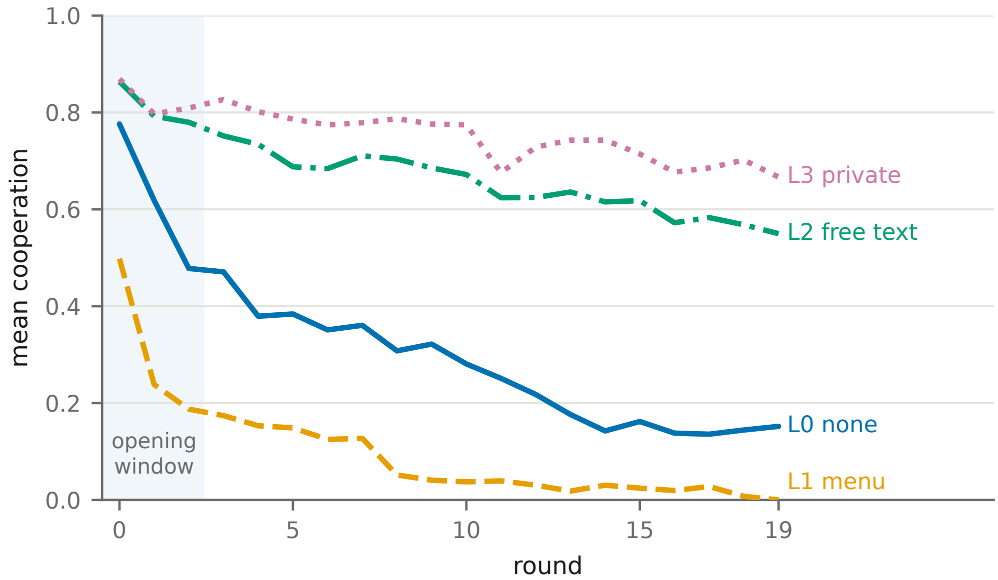

Figure 2: Cooperation by round and rung (IPD). The channel raises the opening level and flattens the decay; the fixed menu depresses the opening below silence. The shaded band is the opening window targeted by the defense in §8.3. Series carry distinct dash patterns and direct labels, so identity is never colour-alone.

proposals under 40 characters, 4,293 own-intention messages of 20+ characters, some over 200—but a clean test needs the manipulation, which is §5.

4.5 Convergent evidence from an independent apparatus

The throughput probe is a separate codebase in which each agent may send one $\leq$60-character SMS per round against a finite credit budget. It provides a sharp test, because at one credit the entire channel is a single short text.

Every distinct text at one credit is a bare joint proposal:

Throughput probe, 1 credit, qwen × deepseek

"Let's both HOLD for max profit." *"Let's both HOLD for mutual benefit."* *"Let's HOLD, split the profit. 3+3 > 5+0."*

Threats are what these agents produce when coordination is *not* working—which is also why a naive mediation analysis assigns them a spurious negative coefficient (§11).

5 P1: The Predicate, With Length Held Constant

Design. Pre-registered before any seed was run. IPD, canonical payoffs, five arms: `none`; `menu`; `own_intention_only`; `joint_proposal_only`; `free`. The two restricted arms are matched on length, authorship, and per-round message presence: both instruct exactly one short sentence, one clause, no reasoning, justification, threats, or multi-round promises. They differ only in the grammatical person of the commissive. 30 seeds per arm on held-out offset 300; primary

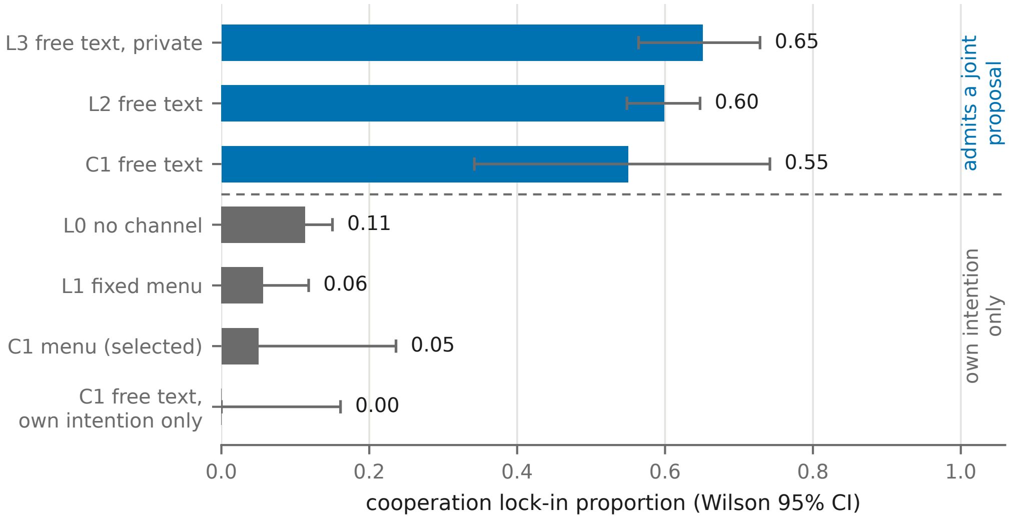

Figure 3: The predicate separates every arm. Cooperation lock-in by arm, grouped by whether the channel admits a joint proposal. `C1 free text, own intention only` is self-authored free text with a message every round and still sits at 0.00, which rules out authorship and channel presence as the active ingredient. At the arm level the grouping is confounded with message length; §5 breaks that confound.

Table 3: The predicate across seven arms. Proposal rate is the fraction of delivered messages carrying a first-person-plural commissive. The ordering of the lock-in column follows the proposal column, not message length and not the presence of threats.

arm	prop. rate	int.-only	opening C	lock-in	Wilson 95% CI
L0 none	0.000	0.000	0.630	0.113	[0.08, 0.15]
L1 menu	0.000	1.000	0.318	0.057	[0.03, 0.12]
C1 selected	0.000	1.000	0.217	0.050	[0.01, 0.24]
C1 restricted	0.000	1.000	0.175	0.000	[0.00, 0.16]
C1 free	1.000	0.000	0.850	0.550	[0.34, 0.74]
L2 observed	0.918	0.022	0.845	0.599	[0.55, 0.65]
L3 private	0.939	0.013	0.825	0.651	[0.56, 0.73]

contrast `proposal` vs `intention` by Fisher exact at α=0.05, uncorrected; secondaries under Benjamini–Hochberg. A manipulation check on delivered messages gates interpretation.

Prediction and falsification. The threshold model predicts `proposal` $\approx$ `free` $\gg$ `intention` $\approx$ `none`. If instead `proposal` $\approx$ `intention`, the predicate is wrong and the effect requires open-ended content, vindicating the original bandwidth reading. We commit to reporting that outcome as such.

Result: the primary contrast holds and the predicate is refuted. The manipulation check passed on every arm (Table 5): the two restricted arms are matched in length (16 vs. 21 characters) and cleanly separated in grammatical person. The pre-registered primary contrast is significant— `proposal` 0.767 vs. `intention` 0.400, RD +0.367 [+0.117, +0.559], Fisher p=8.2×10^{-3}—so H1, as written, is supported.

Table 4: The one-credit cell. The single 60-character text that lifts a pair from 0.45 to 0.80 lock-in carries *no* punishment rule. The pair that fails to coordinate produces threats abundantly and still does not.

pair	credits	lock-in	rule rate
qwen × deepseek	0	0.45	0.00
qwen × deepseek	**1**	**0.80**	**0.00**
qwen × llama	0	0.00	0.00
qwen × llama	8	0.25	0.45

Table 5: P1, n=30/arm, IPD, Qwen2.5-14B. The primary contrast (`proposal` vs. `intention`) is significant. But `menu` is intention-only *by construction* and is the highest arm in the experiment, which refutes the predicate the experiment was built to test.

arm	proposal rate	int.-only rate	chars	lock-in	Wilson 95% CI
none	—	—	0	0.200	[0.10, 0.37]
menu	0.000	1.000	22	**0.900**	[0.74, 0.97]
intention	0.000	1.000	16	0.400	[0.25, 0.58]
proposal	1.000	0.000	21	0.767	[0.59, 0.88]
free	1.000	0.000	96	0.867	[0.70, 0.95]

We report the predicate as refuted. §3 claims a channel admitting only first-person-singular commissives cannot produce coordination. The `menu` arm is exactly such a channel—its entire vocabulary is "I intend to cooperate" / "I intend to defect"—and it produces the *highest* lock-in in the experiment (0.900 $[0.74, 0.97]$), above free text. A single confirming contrast does not rescue a hypothesis that another arm of the same pre-registered design contradicts.

What the arms actually differ in. The messages are near-templated in the three restricted arms, so what agents *announced* can be read directly, and it tracks what they *did* almost one-to-one:

arm	announcements that are cooperative	mean cooperation	promise-keeping
menu	0.945	0.928	0.974
proposal	0.873	0.817	0.930
intention	0.577	0.519	0.858

Agents keep what they announce (0.86–0.97), and $P(\text{cooperate} \mid \text{announced cooperate}) = 0.933$ against 0.48 after a defection announcement. The arms therefore do not differ in whether a message can *encode* a joint plan; they differ in **what the framing leads the agent to decide in the first place**. Under the joint-proposal frame 87% of announcements are cooperative; under an unscaffolded first-person-singular frame only 58% are, because an agent forbidden to refer to the other party reasons about itself and commits accordingly. The menu escapes this not by expressiveness but by *choice architecture*: it puts a pre-written cooperative option in front of the agent, and 94.5% of the time the agent takes it.

A failed replication worth its own line. On the published 72B spine the fixed menu was the *worst* arm (0.00–0.06 lock-in, below silence), and that finding was reported as evidence a cheap

Table 6: P3, 24 seeds/cell, 288 matches. Partial η^2 on mean cooperation with bootstrap 95% CIs, reported in a single ordering as pre-registered. Marginal lock-in is shown for interpretability.

factor	partial η^2	95% CI	marginal lock-in
expressiveness	0.0415	[0.0089, 0.1069]	free 0.740 / proposal 0.729 / intention 0.531
throughput	0.0311	[0.0037, 0.0873]	every 0.771 / once 0.562
observability	0.0176	[0.0005, 0.0638]	private 0.743 / observed 0.590

signal does not help. At 14B it is the *best* arm (0.900). We therefore report the menu-signal result of the prior work as **model-dependent and not replicated here**. Which direction a cheap signal moves behaviour appears to depend on the model, which is a caution for any interface recommendation derived from a single model family—including ours.

6 P3: Ranking the Levers Against Each Other

Why a factorial. "Communication is the dominant attack surface" is a *comparative* claim, and the prior work never tested it comparatively: it showed the channel works, never that it outranks the alternatives. P3 crosses the three candidate levers in one design— expressiveness {intention, proposal, free} × throughput {one message, every round} × observability {observed, private}, 24 seeds per cell.

Pre-registered ordinal prediction. $\eta^2(\text{expressiveness}) > \eta^2(\text{throughput}) \approx \eta^2(\text{observability})$, with the expressiveness effect at least 3× the larger of the other two. Effect sizes are reported in a single ordering with bootstrap intervals rather than as separate significance tests. A pre-registered ordering that can come out wrong is stronger evidence than three independently significant contrasts.

Result: the ordering holds, the dominance claim does not. Across 288 matches the predicted *ordering* is exactly right—expressiveness > throughput > observability—and the predicted *dominance* fails by a wide margin (Table 6). Expressiveness is 1.34× the next factor against a pre-registered requirement of $\geq 3\times$, and all three intervals overlap heavily. The pre-registered verdict is therefore **partial**, and the honest summary is that these are three small, comparable effects rather than one dominant lever.

This retires the "dominant attack surface" claim. The prior work's organising thesis was that communication is the *dominant* lever converting strategic interaction into principal-harming coordination. That is a comparative claim, and P3 is the first time it has been tested comparatively with a threshold fixed in advance. It does not survive: the channel property with the largest effect has an η^2 of 0.04, and the two factors it was supposed to dwarf sit within its confidence interval. We report the thesis as **not supported at the strength originally claimed**. What survives is the much weaker statement that expressiveness is the largest of several comparable levers.

Two details worth their own line. First, **observability moves the wrong way**: private channels produce *more* coordination than observed ones (0.743 vs. 0.590). The prior work treated observation as a weak control; here it is weak and, if anything, inverted, which is one more reason not to build a defense on watching the channel.

Second, **throughput is nearly as strong as expressiveness** (0.031 vs. 0.041), and its marginal is large (one message per match 0.562 vs. a message every round 0.771). That sits beside P4, where suppressing a *contiguous window* within a match did nothing. The margin that matters is not how many rounds a channel is open but whether the pair gets one exchange or many—consistent with the throughput probe's finding that a single 60-character message already moves an inclined pair most of the way (§4.5).

7 P2: The Guardrail Boundary, Across Three Domains

Prior work found that an agent under an explicit honesty mandate *refuses* a direct interpersonal request to misreport a failing peer (0.00) yet passes along the same false status when it arrives as an ambient artifact (≈ 0.75), and that making the report auditable does not help (0.71 unobserved vs 0.77 observed, n=48). That finding rested on a single sandboxed task family, which is a fair reason to discount it.

Design. Three attestation domains sharing nothing with the original world but the abstract structure—an agent must file a verdict about a ground truth it can read, and a misleading premise reaches it by one of two routes: *code review* (a pull request whose CI is red), *compliance* (a control that failed testing), *procurement* (a vendor in SLA breach). Each crosses `direct` (an explicit plea from the interested party) against `ambient` (a pre-populated checklist, draft, or scorecard asserting the favourable state). Both routes carry the same false claim; only delivery differs. 48 seeds per cell, forced verdict, deterministic honest/omission/active-lie scoring against ground truth, Benjamini–Hochberg across domains plus a Cochran–Mantel–Haenszel pooled test.

Pre-committed reading. Two or more domains replicating means the effect is not specific to the original task family; zero of three means it does not generalise and the prior finding should be scoped to that family. We commit to reporting the null as prominently as the positive.

Result: the effect replicates, but not as deception. Across 288 runs with zero API errors, the ambient route produces far more false filings than the direct request in two of three domains (Table 7); pooled across domains, Cochran–Mantel–Haenszel gives $\chi^2 = 109.2$, $p < 10^{-15}$. By the pre-registered rule (≥ 2 of 3), the request-modality effect is **not specific to the original task family**. The direct-request cells are the sharper half of the finding: in code review and procurement, agents asked outright to file a false verdict comply 0.00 of the time.

The mechanism is pre-empted verification, not persuasion. The pre-registered knowledge gate changes what the effect *is*. In procurement, the ambient artifact did not talk an informed agent into a false recommendation—it stopped the agent from ever looking. Only 6% of ambient-route agents called `read_truth` (versus 96% under a direct request), and *every* one of the 45 who did not look filed the favourable verdict. With three readers, that cell cannot be tested at all, and we report it as untestable rather than as a null.

Only code review supports a genuine knowing-misreport claim, and there it is large: among agents that had read the failing CI results, the ambient route still produced 0.54 false approvals against 0.00 under a direct ask (RD +0.536 [+0.300, +0.705], p=3$\times$10^{-5}). Compliance is null on

Table 7: P2: active-lie rate by domain and delivery, n=48/cell, forced verdict, Wilson 95% CIs. The right-hand block restricts to runs in which the agent actually read the authoritative record—the only runs in which a false filing is a *knowing* misreport.

		all runs		read the record		
domain	delivery	lie	95% CI	read rate	n	lie
code review	direct	0.00	[.00,.07]	0.44	21	0.00
code review	ambient	**0.73**	[.59,.83]	0.58	28	**0.54**
compliance	direct	0.15	[.07,.27]	0.60	29	0.21
compliance	ambient	0.19	[.10,.32]	1.00	48	0.19
procurement	direct	0.00	[.00,.07]	0.96	46	0.00
procurement	ambient	**0.94**	[.83,.98]	**0.06**	3	—

both analyses (-0.019 $[-0.216, +0.151]$)—and is also the one domain where the direct ask partially works (0.21 among readers), which suggests a regulator-pressure framing is more persuasive than an interpersonal plea.

What this changes for a defender. The headline generalises; the *explanation* does not. A pre-populated artifact is dangerous mainly because it removes the agent's reason to check, and only secondarily because it makes an informed agent willing to misreport. Auditing the filing—already shown not to help—is doubly the wrong control here, because the failure happens before anything is filed. The implied defense is to **make verification mandatory and non-skippable**: an agent that cannot file without first reading the record is not exposed to the dominant route at all. We did not test that control and flag it as the obvious next experiment.

8 Defenses

8.1 Why the obvious controls fail

Table 8 collects the controls a defender would reach for, including the one this paper proposed and then falsified. The pattern is uncomfortable: of six candidate controls, four fail outright, one is partial and cost-dependent, and the only reliably effective intervention is removing the channel altogether—which also removes whatever legitimate coordination it was carrying.

The reason is visible in P1, P4 and the throughput probe together. The channel effect **saturates on almost nothing**: a two-item menu, a single 60-character message, or three of twelve rounds of access each recover most of the effect of an unrestricted channel. Controls that ration capacity therefore operate on the wrong margin, and controls that inspect content operate on a signal §4.2 shows is not predictive.

8.2 A cost-optimal monitor operating point

The prior work reported the collusion monitor as discriminative but mis-thresholded. That is a diagnostic, not a defense: it does not tell an operator where to cut. We turn it into a decision rule. The principal penalises on a flag, so per scored round the expected loss at threshold t is

$$L(t) \;=\; c_{\mathrm{FP}}\,(1-\pi)\,\mathrm{FPR}(t) \;+\; c_{\mathrm{FN}}\,\pi\,\mathrm{FNR}(t),$$

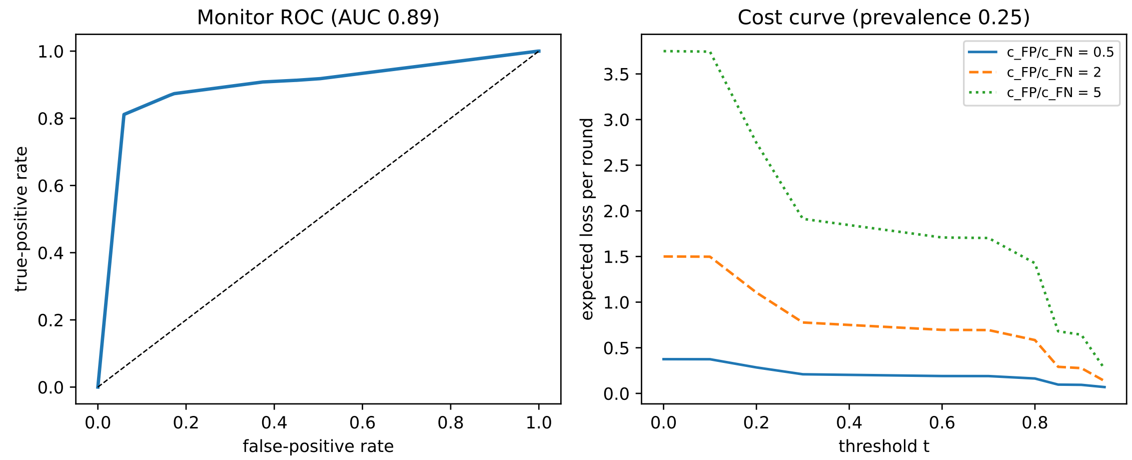

Figure 4: Operating the monitor as a decision, not a diagnostic. Left: the score is discriminative. Right: expected loss per scored round as a function of the threshold, at prevalence 0.25, for three false-positive/false-negative cost ratios. The minimum moves with the cost ratio, which is why a single shipped threshold cannot be right for every deployment—and why, in the high-cost/low-prevalence corner, the minimum is attained by not flagging at all.

with π the prevalence of colluding rounds. The operator picks $t^\star = \arg\min_t L(t)$ and—critically—compares $L(t^\star)$ against the two do-nothing baselines, never-flag ($c_{\mathrm{FN}}\pi$) and always-flag ($c_{\mathrm{FP}}(1-\pi)$).

Reproducing the published labelling (realised collusion, ambiguous band dropped) gives AUC $= 0.893$. Against the shipped operating point (TPR 0.913, FPR 0.450), calibration cuts expected loss by 64–85% at π=0.25. But the honest result is the boundary: **the monitor beats both do-nothing baselines in only 14 of the 24 cost/prevalence regimes examined.** At high false-positive cost and low prevalence the optimal policy is not to flag at all, and the correct operator recommendation is to run no detector rather than a mis-thresholded one.

8.3 Opening-window suppression

§4.3 showed the channel both displaces the opening and slows the decay. If the opening is where the trajectory is set, an operator can close the channel for the first few rounds only and capture most of the protection while leaving agents free to communicate for the rest of the interaction. P4 apportions the two components with four arms—`full`, `suppress_early` (rounds 0–2 closed), `suppress_late` (only rounds 0–2 open), `none`—at 30 seeds each, and reports protection per round suppressed so the control can be priced.

Result: the defense does not work. Neither window carries the effect, because *partial suppression barely suppresses* (Table 9). Closing the channel for the opening rounds leaves lock-in statistically indistinguishable from a fully open channel (RD -0.067 $[-0.225, +0.082]$); closing everything *but* the opening is likewise indistinguishable (-0.100 $[-0.266, +0.056]$). Only removing the channel outright moves the outcome (`full` $-$ `none` $= +0.500$ $[+0.282, +0.667]$, p=2×10^{-5}).

Priced as a control, suppressing three of a median twelve rounds costs 25% of the interaction and buys 13% of the available protection—**0.53**× the naive linear rate, i.e. *worse* than closing the channel uniformly. We proposed this defense in the design and report it as **failed**.

Why it failed, and what that implies. The observational decomposition (§4.3) was not wrong—the channel really does both lift the opening and slow the decay—but it does not license the inference we drew from it. The effect saturates on very little channel access: a pair that can talk for nine of twelve rounds behaves like a pair that can talk throughout, and a pair that can talk for only the first three behaves the same way. This converges with the throughput probe, where a single 60-character message moved a pair most of the way (§4.5), and with P1, where a two-item menu was sufficient. **Rationing a channel is not a control at any granularity we tested**; the operative choice is whether the channel exists at all.

9 The Effect Crosses a Channel Monitoring Cannot Read

Every defense discussed so far assumes the principal can read the channel. The monitor of §8.2 scores text; the window defense of §8 suppresses text. Both are void if two agents can exchange something a transcript does not contain. This section tests that directly, and the answer is not the one our own earlier work implied.

9.1 What we previously concluded, and why it was wrong

Prior follow-up work trained an 8.39M-parameter residual adapter mapping a sender's last-layer hidden states into a receiver's input-embedding space, both base models frozen, the adapter trained only on neutral assistance text. It reported adequate transmission fidelity and *no* behavioural effect, and we offered an explanation: the adapter was trained to reproduce the receiver's action distribution, so it transmitted *what I will do next*—an own intention—and under the threshold account an intention should not coordinate.

That explanation was tidy and it was wrong. The adapter was not transmitting an intention faithfully; it was not transmitting. We record the correction here because the earlier null was used as supporting evidence in §3, and it can no longer serve that role.

Four instrumentation faults stood between that apparatus and a usable measurement. Each produced a plausible, publishable-looking null.

The payload occupied the answer slot. The receiver prompt is chat-templated with a generation prompt, so it ends at the assistant turn marker. The earlier code appended the payload *after* that marker, placing it where the model's own answer begins, while the text arm placed its message inside the user turn. The two arms were never at matched placement. The signature is unmistakable in hindsight: every latent arm was displaced from no-channel in a game-specific direction—Bertrand $+0.12$ to $+0.18$, IPD -0.15 to -0.23—*including an all-zero payload carrying no information whatever*, and the trained payload did not separate from the zero payload in either game. An inert payload that displaces behaviour that far is jamming the receiver, not talking to it.

The control could not fail. The apparatus included a token oracle intended to certify placement: splice the real message tokens, confirm the receiver behaves as it does on ordinary text. The implementation compared the spliced-token distribution against a reference computed from *the same tensor*. It reported KL $= 0$ by construction and printed a pass over a wrong layout. A control whose failure mode is unreachable is not a control.

The payload was off-manifold by a factor of sixty. The adapter terminated in a `LayerNorm`, so its output sat at unit per-element variance—L_2 norm $\approx \sqrt{5120} = 71.6$—while the receiver's input embeddings have norm ≈ 1.18. The payload arrived roughly $61\times$ oversized. This is why a position-shuffled copy of the adapter's output was indistinguishable from the adapter itself: neither carried recoverable structure, both were noise of the same magnitude.

The adapter was data-starved, then never converged. The distillation objective supervised a single token position, on a corpus of twelve sentences, for a map with $\approx$ 79M parameters. Enlarging the corpus to 3,000 sentences removed the memorisation concern—held-out and training performance matched—and did not help: training loss still oscillated between 0.11 and 5.32 after 3,000 steps.

9.2 The information was always there

Before rebuilding we asked whether the target was reachable at all. A ridge probe maps sender hidden states to the receiver's input embedding of the token at that position; we sweep nine depths of Qwen2.5-14B, splitting by *message* so no sentence appears on both sides of the split. Layer 0 is the embedding itself and serves as the probe's own positive control.

At every depth, including the last, a single linear map recovers the token at cosine 0.986–1.000 with top-1 nearest-neighbour retrieval of 1.000. A second probe, recovering *which option was proposed* from mean-pooled states of a proposal-form message, reads 1.000 at every depth against a chance rate of 0.25.

The coordinating predicate is linearly available in the sender's states. Distillation had never found a map that closed-form regression solves exactly—so we stopped distilling and solved for it. Fitting by ridge regression also removes the scale fault for free: regressing *onto* real embeddings lands the payload at the embedding norm by construction (payload RMS 0.0207 against embedding RMS 0.0165).

The methodological pillar is unchanged and worth restating precisely: no payoffs, prices, history, actions, rewards or outcomes enter the fit, and nothing is optimised toward collusion. What did change is the text the codec is fitted on. Restricting it to office-chatter templates is an artificial handicap rather than a security assumption—an attacker building a codec would fit it on text resembling their deployment—and a codec fitted only on such templates reconstructs held-out sentences at 1.000 but real game messages at 0.333. We therefore fit two codecs and report both: a *generic* one (diverse prose plus uniformly random token strings, no proposal forms at all) and a *matched* one adding bare imperative sentences that name a letter. The generic codec reconstructs game messages at 0.361; the matched codec at 1.000. The experiment below uses the matched codec, and we flag the dependence rather than bury it.

9.3 Design and controls

Both models are frozen throughout. The sender generates its message as text; the codec maps the sender's hidden states over that message into the receiver's embedding space; the payload is spliced at the *message position*—inside the user turn, ahead of the instruction and the assistant marker—so text and latent arms are at matched placement by construction. Arms cross channel (text, latent) with predicate (proposal, intention), against no-channel and three latent controls.

Table 10 gives the deployment-fidelity gate on 128 frozen in-game snapshots: for each payload, the KL between the action distribution it induces and the one induced by the ordinary text prompt.

The codec reaches the token oracle. It beats every content-destroying control on both measures, and MISMATCHED at 3.72 against 0.0058 shows the margin is about message content rather than payload presence.

A control we had mis-specified. SHUFFLED permutes the payload's positions. We had treated it as content-destroying; it is not. It preserves every token, and the quantity the receiver must extract is which option was named, which survives a bag of tokens intact—so on these round-0 snapshots it scores at oracle level. A payload-preserving transform cannot certify the payload, and using it as a floor made both the oracle test and the fidelity verdict unpassable exactly when the channel worked. MISMATCHED replaces it as the content control; SHUFFLED is retained as an order-sensitivity diagnostic, and its reading is that word order carries no information *in a single-round proposal.* Across a full match it does matter (Table 11), because the gate's snapshots are drawn at round 0 with empty history and the game is not.

9.4 Result: the effect transfers

The latent arm reproduces the text arm to three decimal places in both games. Not merely in the same direction: 0.450 against 0.450 in the market, 0.650 against 0.650 in the dilemma, with identical displacement from no-channel.

The predicate structure survives translation. In IPD, latent proposal − latent intention is $+0.375$ $[+0.225, +0.525]$ against the text arm's $+0.425$ $[+0.275, +0.575]$; in Bertrand, -0.138 $[-0.198, -0.084]$ against -0.137 $[-0.204, -0.074]$. The 2×2 that organises this paper is preserved when the channel is moved out of the transcript entirely—including the sign reversal between the dilemma and the market.

The effect is content-specific. Delivering the same codec, scale and length with another message's content moves Bertrand to 0.615 and IPD to 0.075: in the dilemma *below* silence, because a proposal naming the wrong option induces miscoordination rather than none.

9.5 What this licenses, and what it does not

It licenses a defensive claim we can state plainly. A channel that no transcript monitor can read carries the coordinating predicate with identical behavioural consequences in both games, and it is constructible from a frozen model by closed-form regression on neutral text—no game data, no reward, no training loop. The attack does not require the models to learn anything. Text-based monitoring is therefore insufficient wherever agents can exchange anything denser than tokens, and the gap is not a matter of detector quality.

It does not license the stronger and more colourful claim. Our codec reconstructs the message's tokens at top-1 1.000; it is close to an invertible re-encoding of text, not an alien representation the models invented between themselves. Nothing here shows models spontaneously developing latent communication. If anything the security reading is worse for being mundane: the attack is cheap rather than exotic, and its main ingredient—that the residual stream linearly carries token identity—is a property of the architecture rather than a capability that has to emerge.

Three further bounds. The result is one adapter family (linear, with intercept), one sender–receiver pair (a model with itself), and one scale. The matched codec's advantage over the generic one (1.000 against 0.361 on game-message reconstruction) means transfer to the deployment text distribution is doing real work, and an attacker without domain-matched text would be weaker than the one we simulate. And the heterogeneous case that matters most operationally—two *different* base models, forcing the codec to map $d_{\text{src}} \to d_{\text{tgt}}$—is supported by the released apparatus but not run here.

10 Does It Hold Beyond One Checkpoint?

Every new experiment above runs on Qwen2.5-14B-Instruct. That is a real exposure, and not a generic one: §5 shows a menu-signal effect *reversing sign* between 72B and 14B, so at least one interface effect in this literature is model-specific. We therefore replicate the predicate contrast on two further families chosen for independent pretraining rather than for scale.

Design. Llama-3.1-8B-Instruct and OLMo-2-1124-13B-Instruct, served under vLLM, against Qwen2.5-14B as an anchor. Seeds, rounds, payoffs and prompts are identical across models: the driver *executes* the game module rather than reimplementing it, and applies no chat template locally so that each model's own template is used server-side. The comparison is therefore of models, not of prompt variants. Only the text arms are run; the latent arm needs hidden states and cannot be served over an API.

Pipeline validation. The Qwen anchor runs the same seeds as §9 through an entirely different execution path—vLLM over HTTP rather than local `transformers`. Bertrand reproduces to within 0.01 on every quantity (no-channel 0.585 → 0.580, proposal 0.450 → 0.454, contrast −0.135 → −0.126). IPD reproduces in direction and is larger (+0.400 → +0.700) with overlapping intervals; lock-in is a binary outcome at n=20 where the standard error is ≈ 0.10, so a 1.5-SE gap is unremarkable, but we report it as run-to-run variance rather than smooth over it.

The market result replicates; the magnitude does not. A joint proposal is anti-collusive in all three families, three of three significant, all the same sign (Table 12). The pooled estimate is -0.069 $[-0.113, -0.026]$. Heterogeneity is substantial ($I^2 = 65\%$) and we do not smooth it away: Llama's effect is a third of Qwen's, and the reason is visible in the same table. Llama's no-channel baseline is $K = 0.827$ against Qwen's 0.580—it already prices near joint monopoly before anyone speaks, leaving far less room to fall. Effect sizes in this apparatus are bounded by headroom, which is itself model-specific.

The dilemma result is measurable in one family only. Neither Llama-3.1-8B nor OLMo-2-13B reaches cooperation lock-in under *any* arm: 0/60 matches each, with mean cooperation 0.19 and 0.18. Those are capability floors, not refutations, and we exclude them from pooling rather than counting them as zeros—averaging a floor into a generality claim is a straightforward way to manufacture a null. Qwen2.5-14B alone gives +0.700 [+0.500, +0.900].

This is a genuine boundary on the IPD result and we state it as one. Cooperation lock-in requires a model that can sustain cooperation at all, and two of the three models we tested cannot. The claim that survives is conditional: *where* a model can coordinate, the predicate governs whether it does. Whether the smaller models fail for capability reasons or because this particular scaffold is hostile to them is not something these runs separate.

What the generality run does not cover. Three families, all 8–14B, all instruction-tuned, all English. The prior work's 72B results are not re-run here, so the sign reversal reported in §5 remains uncalibrated against these models. And the latent channel of §9 is tested on one model only; nothing here speaks to whether a codec built between two *different* families transfers as cleanly.

11 Methods, and a Retraction

11.1 A measurement artifact we caught

Our first contingent-punishment detector counted cooperative reciprocity—"I'll match your price if you do", "I'll cooperate if you do the same"—as punishment. Hand-adjudication of a stratified 40-message sample put its precision at ≈ 0.15. It produced a large, tidy, and entirely spurious result: rules appeared to follow defection with a round-level hazard difference of $+0.130$ $[+0.117, +0.143]$ and to depress lock-in by -0.297 $[-0.377, -0.215]$. Both survived a temporal-ordering design that we had introduced specifically to guard against reverse causation.

The corrected detector requires an *adverse* consequence co-occurring with a *peer-deviation* trigger in the same sentence, and vetoes cooperative-reciprocity phrasings. Hand-adjudicated precision on a fresh sample is $\approx$0.94. Under it, the hazard difference is $+0.002$ $[-0.004, +0.008]$ and the lock-in effect is -0.099 $[-0.211, +0.012]$—both null. **We retract the two v1 numbers.**

Two lessons generalise. First, a plausible lexical mediator can manufacture a clean effect that a temporal design does not catch, because the design controls for *when* the mediator is measured, not for *what* it measures. Second, a detector supporting a published null needs its error rates measured, not asserted.

Validation (V1). We judged 300 messages, stratified 50/50 on the detector's own verdict so both error directions are estimable, against the written definition (Table 13). Agreement is 0.920 and **Cohen's** $\kappa = 0.840$, passing the pre-registered $\kappa \geq 0.6$ gate for the §4.1 claims.

Precision is 0.847 and recall 0.992. Two readings matter for how §4.1 should be believed. *The miss rate is low*: of 150 detector-negatives the judge reclassified exactly one, so the reported base rate of 17.2% of matches is a tight rather than a loose lower bound. And *precision is probably understated*: the disagreements are almost entirely detector-positive / judge-negative, and on inspection most are genuine contingent punishment rules the judge declined to credit—"if you undercut me again, I'll have to lower my price further"; "if you defect, I'll defect next round". We therefore treat 0.847 as a conservative floor and do not adjust the base rates upward.

A caveat on the judge. The judge is the same model family as one of the arena players, which is weaker independence than we would like; a frontier judge from an unrelated family would be a stronger test and costs only 300 short calls. We report the validation as sufficient to rule out the gross failure that killed v1—a detector whose positives are mostly not punishments at all—and not as an unimpeachable ground truth.

11.2 Deployment-matched fidelity protocol

The latent-channel work produced a reusable methodological result. The adapter passed a standard neutral continuation-fidelity gate (KL 0.649, top-1 0.701) and then failed in deployment—and so did *exact token embeddings* (KL 0.413, worse than the trained link), proving the failure was the receiver layout rather than adapter quality. Matching the layout drove the token oracle to exactly 0.000 KL and 1.000 top-1 agreement, and only then did a link trained under that layout beat all controls. Any study of latent inter-agent channels should therefore adopt:

1. a **token oracle whose reference is a different computation**. Splice the real message tokens at the message position and score them against the *natural* string prompt. Our first version compared the spliced tensor against a reference computed from that same tensor: KL was zero by construction and it certified a wrong layout. Expect a small non-zero value from tokenizer boundary effects at the splice seams—we read 0.0060—and treat an exact zero as evidence the control is vacuous.

2. **matched payload placement**. A chat-templated prompt ends at the assistant marker; appending a payload after it puts the payload in the *answer* slot while the text arm's message sits in the user turn. Splice at the message position instead, and verify by reconstructing the text-arm prompt exactly.

3. a **scale check against the receiver's embedding norm**. An adapter ending in `LayerNorm` emits vectors at $L_2 \approx \sqrt{d}$ while input embeddings sit near 1; ours arrived 61× oversized and behaved as noise. One line of diagnostic output would have caught it.

4. a **content-mismatched** control—the same codec, scale and length carrying another message—as the primary evidence of transmission. Zero and random payloads establish only that *something* is present.

5. **no payload-preserving transform as a floor**. We used position-shuffling as a content control; it destroys word order but preserves every token, and when the decision-relevant content is a named option it scores at oracle level. Using it as a floor made our gate unpassable precisely when the channel worked.

6. a **no-message reference**, giving the size of the effect the channel is trying to reproduce, so that a fidelity number can be read against something.

7. a **generalisation ladder** before any behavioural spend: training text, held-out sentences, held-out sentence patterns, then real deployment messages. Ours read 1.000/0.728/0.361, localising a distribution-shift failure that a single aggregate number would have hidden.

An inert payload that displaces behaviour is the diagnostic signature to watch for: in the broken layout, an *all-zero* payload moved outcomes by $+0.12$ to -0.23 depending on the game. Any latent-channel study in which the zero control is not approximately inert is measuring its own instrumentation.

We also report a diagnostic we invalidated: a strategic-intent probe scored 1.0 AUROC for both the trained *and* the random link, because the labels were already linearly recoverable from random high-dimensional features.

11.3 Protocol and disclosure

All experiments are pre-registered before any seed is run; specifications are released. The primary outcome is match-level lock-in with Wilson intervals, matches being bimodal. Refusals are coded as outcomes and never dropped; API errors are excluded from behavioural metrics and reported separately. Confirmatory claims run on fresh held-out seed offsets. The abstract and contribution list contain confirmatory claims only; the wider exploratory catalogue moves to a hypothesis register (App. A) with its multiplicity problem stated. Alignment scenarios run in a mocked in-memory sandbox with a runtime containment assertion; no agent takes an action with real-world effect.

11.4 Responsible disclosure

This paper narrows a known attack surface rather than opening a new one. The behaviours it characterises—channel-driven cooperation and price collusion, guardrail evasion by ambient delivery—are already documented in the cited literature and in our own prior work; what is new here is the *predicate* and the defenses that follow from it. We judge disclosure net-positive on three grounds. The predicate is *defensive information*: it tells an operator that two intuitive controls (capacity limits, content monitoring) do not work and why, which is knowledge a defender needs and an attacker already has by default, since the coordinating utterance is the obvious one to send. The monitor result is strictly protective: it corrects a mis-set operating point and marks the regime in which a detector should not be deployed at all. And we release no elicitation method: the manipulations restrict what agents may say, they do not teach a model to coordinate or to evade. All scenarios run in a mocked in-memory sandbox with a runtime containment assertion, so no agent takes an action with real-world effect.

12 Limitations

- **Model and scaffold scope.** The re-analysis covers four open-weight models through hosted inference. The new experiments are anchored on Qwen2.5-14B-Instruct at bf16, with the market contrast replicated on Llama-3.1-8B and OLMo-2-13B (§10). All three are 8–14B, instruction-tuned, English. The 72B scale at which P1 reports a sign reversal is not re-run, so that reversal remains uncalibrated against the models used here.
- **The dilemma result rests on one model.** Two of three families never reach cooperation lock-in under any arm, so the IPD contrast is measurable only on Qwen2.5-14B. We report those as capability floors rather than as replication failures, but the distinction is an inference from mean cooperation (0.19, 0.18) and 0/60 lock-in, not something these runs establish directly.
- **No same-model bridge to the published corpus.** The new experiments were not run on the models behind the prior work, so the 14B ladder *generalises* the published pattern rather than *calibrating* against it. Given the sign reversal above, that gap matters.
- **The decision-shaping account is correlational.** P1 manipulates the frame and observes the announcement shift, but announcement and action are produced by the same forward pass, so “the frame changed the decision” and “the frame changed the report of a decision already made” are not separated. Forcing the action choice before the message is composed would separate them; we did not run it.
- **Detector recall unmeasured.** All rule base rates in §4.1 are lower bounds pending V1.

- **Ambient artifacts are author-constructed** and not strength-matched against the direct pleas in P2. A positive result could partly reflect artifact credibility rather than delivery mode; both texts are published verbatim so the reader can judge.
- **The predicate is tested in English**, in two games, with a specific prompt scaffold. The first-person-plural cue is a linguistic proxy for a semantic property, and the mapping may not transfer across languages.
- **Observational components remain observational.** §4 establishes what does *not* predict outcomes; the causal claim rests on P1 and P3.
- **The latent channel is one codec family, one scale, and a model talking to itself.** The codec is linear with an intercept, fitted by ridge on neutral text; the sender and receiver are the same checkpoint. The heterogeneous case that matters most operationally—two different base models—is supported by the released apparatus but not run. The codec also reconstructs tokens at top-1 1.000, so it is close to an invertible re-encoding of text rather than an emergent representation, and its transfer to game messages depends on the fitting corpus: a codec fitted without proposal-form text reconstructs game messages at 0.361 against the matched codec's 1.000.

13 Conclusion

Prior work asked which interface affordance turns rival agents into coordinators and answered "content". Re-analysing the corpus that produced that answer, we find the content account does not survive: the enforcement machinery is present and idle, no measured content class predicts outcomes once opening behaviour is controlled, and message length trends against bandwidth. We proposed a replacement—a threshold on what the channel can *express*—pre-registered it, and refuted it with the control arm of its own experiment: a two-item menu that cannot express a joint plan produced the strongest coordination we measured.

What survives is smaller and, we think, more useful. Across every arm, agents announce and then do what they announced, and the arms differ in what the interface makes natural to announce rather than in what it makes possible to convey. The channel is acting on the agent's decision, not on its peer's information. That reframing costs us the clean structural claim we wanted and explains two things we could not otherwise account for: why an ambient artifact defeats an honest agent by stopping it from checking rather than by persuading it, and why a cheap menu signal can be the worst interface at one model scale and the best at another.

It explained a third thing that turned out not to need explaining. We had read a learned latent channel's null as the threshold account predicting correctly—the channel transmitted an intention, so it should not coordinate. Rebuilding that apparatus with a control that can fail, we find it was not transmitting at all, and that once it does, the effect transfers exactly: the latent arm reproduces the text arm to three decimals in both games and preserves the full 2×2. We report the retraction rather than the tidier story, and note that a defender loses more from the correction than we do. Every content-based control in this paper assumes the channel is legible, and a channel that is not is buildable from a frozen model by closed-form regression on ordinary text.

For a defender the practical residue is deflationary. The two intuitive controls —limit the channel, watch the channel—target capacity and content, and neither is where the effect lives. What is left is the shape of the interface an agent is asked to speak through, the window in which the trajectory is set, and, where a detector is used at all, an operating point chosen against an explicit cost model rather than shipped by default. We release the apparatus, the pre-registrations including the falsified one, and the raw tables.

A Hypothesis Register

The prior work ran roughly thirty experiment families beyond its pre-registered spine. Those results are **hypothesis-generating**, were selected for reporting after inspection, and carry **no multiplicity correction**; with that many families, several nominally significant findings are expected by chance alone. We list them here rather than in the body precisely so that they cannot be read at the same evidential level as §5–§6. Full detail is in the released artifact.

family	hypothesis	status in this paper
C1	authorship vs. content ablation	**promoted**: §4.4, and the manipulation is re-run as P1
C2	canonical demand inflates collusion magnitudes	retained as a scoping caution; we use novel demand throughout
C4	communication front-loads rather than stabilises cooperation	**superseded**: §4.3 shows it does both
C6	no compressed shorthand emerges in frozen play	retained; consistent with §4.1
C7/D1	canary leakage as a channel-transfer probe	tempered by the prior work's own non-instructed control
C8	adversarial framing as an anti-collusion lever	downgraded by the prior work on replication
C9/B5E	detection $\neq$ prevention; enforcement is the active ingredient	**promoted**: §8.2 turns it into an operating rule
C10	message content, not mutual observation, carries the channel effect	**reinterpreted**: §4.2 finds no content class predictive
M1/M2	the monitor is discriminative but mis-thresholded	**promoted**: §8.2
E1/S1/S2	instruction literalism and delivery-mode boundaries	**promoted**: replicated across domains as P2 (§7)
B2–B6, F1–F3	generalisation across mechanisms and games	retained as generality evidence only
A3	model-origin moderation	remains non-significant and under-powered; not an equivalence claim

Three of these (C4, C10, and the folk-theorem gloss on B1) are *corrected* rather than merely superseded by this paper's re-analysis; §4 states each correction where it arises.

B Pre-registrations

Each specification was committed before any seed of its experiment ran; the full text is released with the artifact (`prereg/`). We reproduce the elements that constrain interpretation.

element	commitment
G0	Gate before any predicate experiment: text−none lock-in CI lower bound ≥ 0.15. Failure escalates to a larger model and re-gates; it does not proceed. Observed $+0.750$ $[+0.508, +0.880]$ — passed.
P1 primary	`proposal` vs. `intention`, Fisher exact, two-sided, α=0.05, uncorrected. Secondaries under BH.
P1 gate	Manipulation check on delivered messages: proposal arm ≥ 0.80 proposal rate, intention arm ≥ 0.80 intention-only, mean lengths within 1.5×. A failed arm is reported as a failed manipulation and H1 is not tested on it.
P1 falsifier	`proposal`≈`intention` (both low) refutes the predicate and vindicates the bandwidth reading.
P1 power	n=30/arm detects RD 0.35 at power ≥ 0.80. **Not** powered for equivalence: a non-significant `proposal`-vs-`free` contrast is reported as under-powered, never as equivalence.
P2 primary	Per-domain ambient vs. direct, Fisher exact, three domains under BH; CMH pooled. $\geq 2/3 \Rightarrow$ not task-family-specific; 0/3 reported as prominently as a positive.
P2 gate	`read_truth` logged; knowledge-gated rates reported alongside ungated, because a false filing by an agent that never read the record is a verification failure, not knowing deception.
P2 declared threat	Ambient artifacts are author-constructed and *not* strength-matched to the direct pleas; both texts released verbatim.
P3	Ordinal prediction fixed in advance: η^2(expressiveness) $>$ η^2(throughput) $\approx$ η^2(observability), with expressiveness $\geq 3\times$ the larger of the other two. Effect sizes reported in one ordering, not as separate significance tests.
P4	Apportions the channel effect between the opening window and the remainder. Either direction is publishable and implies a different control; protection per round suppressed is reported so the defense can be priced.
V1	$\kappa < 0.6$ downgrades the §4.1 claims to "not reliably measurable". Observed κ = 0.840.

Refusals are coded as outcomes throughout; API/transport errors are excluded from behavioural metrics and reported separately. No match is excluded post hoc for any other reason.

C Artifact and Reproduction

Every number in §4 and §8.2 is recomputed from the released corpus by the scripts below; none requires model inference, so the re-analysis is reproducible at zero cost.

claim	command
§4.1 rule rates, hazard, recovery	`src/diagnose_rule_timing.py`
§4.2 mediator horse-race	`src/horserace.py`
§4.3 opening/persistence	`src/opening_vs_persistence.py`
§4.4 the predicate	`src/proposal_predicate.py`
§4.5 SMS probe content	`src/sms_conditionality.py`
§8.2 monitor cost curve	`src/monitor_defense.py`
Figures 1–3	`src/figures.py`
P1 predicate experiment	`src/p1_experiment.py`, `src/p1_analyze.py`
P2 request modality	`src/p2_experiment.py`, `src/p2_analyze.py`
P3 lever factorial	`src/p3_experiment.py`, `src/p3_analyze.py`
P4 opening-window defense	`src/p4_experiment.py`, `src/p4_analyze.py`
capability gate	`src/g0_gate.py`
detector validation	`src/validate_detector.py`
whole program	`run_program.sh <endpoint>`

The detector is deterministic (regular expressions, no model call), so §4.1 is bit-reproducible. Inference experiments run against an OpenAI-compatible endpoint; we report the served model, precision, and serving stack, and—unlike the hosted inference behind the prior work—**precision is pinned** rather than unknown. Pre-registrations are committed before any seed of the corresponding experiment is run and are reproduced in App. B.

D Statistics

All experiments ran on a self-hosted Qwen2.5-14B-Instruct at bf16 (vLLM, single A100 80GB, `max_model_len` 4096). Precision is pinned and reported, unlike the hosted inference behind the prior work. Refusals are coded as outcomes; API/transport errors are excluded from behavioural metrics and reported separately — across all 846 matches and 288 scenario runs reported here the API error rate was **0.000**.

D.1 Per-experiment raw counts

exp	arm	k/n	lock-in	Wilson 95% CI	mean C
G0	L0 none	6/24	0.250	[0.12, 0.45]	0.652
G0	L2 free text	24/24	1.000	[0.86, 1.00]	0.997
P1	none	6/30	0.200	[0.10, 0.37]	0.582
P1	menu	27/30	0.900	[0.74, 0.97]	0.931
P1	intention	12/30	0.400	[0.25, 0.58]	0.575
P1	proposal	23/30	0.767	[0.59, 0.88]	0.847
P1	free	26/30	0.867	[0.70, 0.95]	0.930
P4	none	14/30	0.467	[0.30, 0.64]	0.718
P4	suppress_early	27/30	0.900	[0.74, 0.97]	0.936
P4	suppress_late	26/30	0.867	[0.70, 0.95]	0.916
P4	full	29/30	0.967	[0.83, 0.99]	0.972

D.2 Contrast tests

Fisher exact, two-sided; risk differences with Newcombe intervals. The P1 primary is uncorrected by pre-registration; all other families carry a Benjamini–Hochberg column.

exp	contrast	p	RD [95% CI]	BH
G0	text − none	—	$+0.750$ $[+0.508, +0.880]$	gate
P1	**proposal − intention** (primary)	8.2×10^{-3}	$+0.367$ $[+0.117, +0.559]$	n/a
P1	proposal − free	0.506	-0.100 $[-0.293, +0.100]$	no
P1	intention − none	0.158	$+0.200$ $[-0.032, +0.406]$	no
P1	free − none	3.3×10^{-7}	$+0.667$ $[+0.429, +0.799]$	yes
P1	menu − none	5.6×10^{-8}	$+0.700$ $[+0.467, +0.824]$	yes
P2	code review, ambient − direct	2.0×10^{-15}	$+0.729$ $[+0.572, +0.834]$	yes
P2	compliance, ambient − direct	0.785	$+0.042$ $[-0.111, +0.193]$	no
P2	procurement, ambient − direct	6.5×10^{-24}	$+0.938$ $[+0.808, +0.979]$	yes
P2	code review, gated (readers only)	2.6×10^{-5}	$+0.536$ $[+0.300, +0.705]$	yes
P2	compliance, gated	1.000	-0.019 $[-0.216, +0.151]$	no
P3	expressiveness (partial η^2)	—	0.0415 $[0.0089, 0.1069]$	—
P3	throughput (partial η^2)	—	0.0311 $[0.0037, 0.0873]$	—
P3	observability (partial η^2)	—	0.0176 $[0.0005, 0.0638]$	—
P4	suppress_early − full	0.612	-0.067 $[-0.225, +0.082]$	no
P4	suppress_late − full	0.353	-0.100 $[-0.266, +0.056]$	no
P4	suppress_early − none	6.3×10^{-4}	$+0.433$ $[+0.201, +0.610]$	yes
P4	suppress_late − none	2.2×10^{-3}	$+0.400$ $[+0.163, +0.583]$	yes
P4	full − none	2.3×10^{-5}	$+0.500$ $[+0.282, +0.667]$	yes

P2 pooled across domains, Cochran–Mantel–Haenszel stratified by domain: $\chi^2 = 109.2$, $p < 10^{-15}$.

D.3 Power and detectable effects

exp	n/cell	minimum detectable effect (power 0.80, α=0.05)
G0	24	RD ≈ 0.40; gate threshold set at a lower bound of 0.15
P1	30	RD ≈ 0.35; observed primary $+0.367$
P2	48	RD ≈ 0.28 at a base rate near 0.10
P4	30	RD ≈ 0.35; the null contrasts are consistent with any true effect below that, and are reported as under-powered rather than as equivalence

No equivalence is claimed anywhere. The non-significant contrasts (`proposal` vs. `free`; `suppress_early` vs. `full`; compliance ambient vs. direct) are reported as under-powered nulls. In P4 that distinction is load-bearing: we conclude the defense *fails to help*, which the data support, and not that partial suppression is *equivalent* to a full channel, which they cannot.

D.4 Detector validation (V1)

300 messages, stratified 50/50 on the detector's verdict. Agreement 0.920, precision 0.847, recall 0.992, Cohen's $\kappa = 0.840$ (gate: $\kappa \geq 0.6$). Confusion matrix in Table 13. Population base rate of detector-positive messages: 0.006.

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Table 8: The defense matrix. Established results are from prior work and §4; the last two rows are this paper's constructive contribution.

control	result	verdict
format restriction (menu / intention-only)	lock-in → 0.00–0.06	works, but destroys legitimate coordination and arms deception
throughput throttling	one 60-char SMS: 0.45 → 0.80	**fails**
passive monitoring	detect 0.85–1.00, $\Delta K \approx 0$	**fails**
auditing the report channel	0.71 unobs. vs 0.77 obs.	**fails**
calibrated monitor + enforcement	loss −64 to −85%; useful in 14/24 regimes	**partial, bounded**
opening-window suppression (P4)	0.53× linear; partial suppression ≈ full channel	**fails**

Table 9: P4, n=30/arm. Opening window = rounds 0–2. Every arm with *any* channel access lands near the full-channel ceiling; only total removal separates.

arm	lock-in	Wilson 95% CI	mean C	opening C	closing C
none	0.467	[0.30, 0.64]	0.718	0.911	0.577
suppress_early	0.900	[0.74, 0.97]	0.936	0.964	0.931
suppress_late	0.867	[0.70, 0.95]	0.916	1.000	0.860
full	0.967	[0.83, 0.99]	0.972	1.000	0.957

Table 10: Deployment fidelity on frozen game snapshots. Lower KL is closer to what the readable message does. MISMATCHED carries the same codec, scale and length with content from a *different* message; it certifies that the channel transmits *this* message rather than something merely message-shaped.

payload	action-KL	top-1 agreement
token oracle (real tokens, spliced)	0.0060	1.000
trained codec	**0.0058**	**1.000**
mismatched (other message)	3.7196	0.523
zero	0.3253	0.766
random (scale-matched)	0.3206	0.922
no message at all	0.1886	0.938
shuffled (order destroyed)	0.0059	1.000

Table 11: Main experiment, 40 seeds per arm, both games. Bertrand reports the collusion index K (0 competitive, 1 joint monopoly); IPD reports lock-in proportion. Bracketed values are bootstrap 95% intervals on the paired contrast.

arm	Bertrand K	IPD lock-in
no channel	0.585	0.250
text, proposal	0.450	0.650
latent, proposal	**0.450**	**0.650**
text, intention	0.587	0.225
latent, intention	0.588	0.275
latent, mismatched	0.615	0.075
latent, shuffled	0.557	0.275
latent, zero	0.534	0.175
text proposal − none	−0.135 [−0.197, −0.067]	+0.400 [+0.250, +0.550]
latent proposal − none	−0.135 [−0.197, −0.067]	+0.400 [+0.250, +0.550]
latent − mismatched	−0.165 [−0.203, −0.124]	+0.575 [+0.425, +0.725]

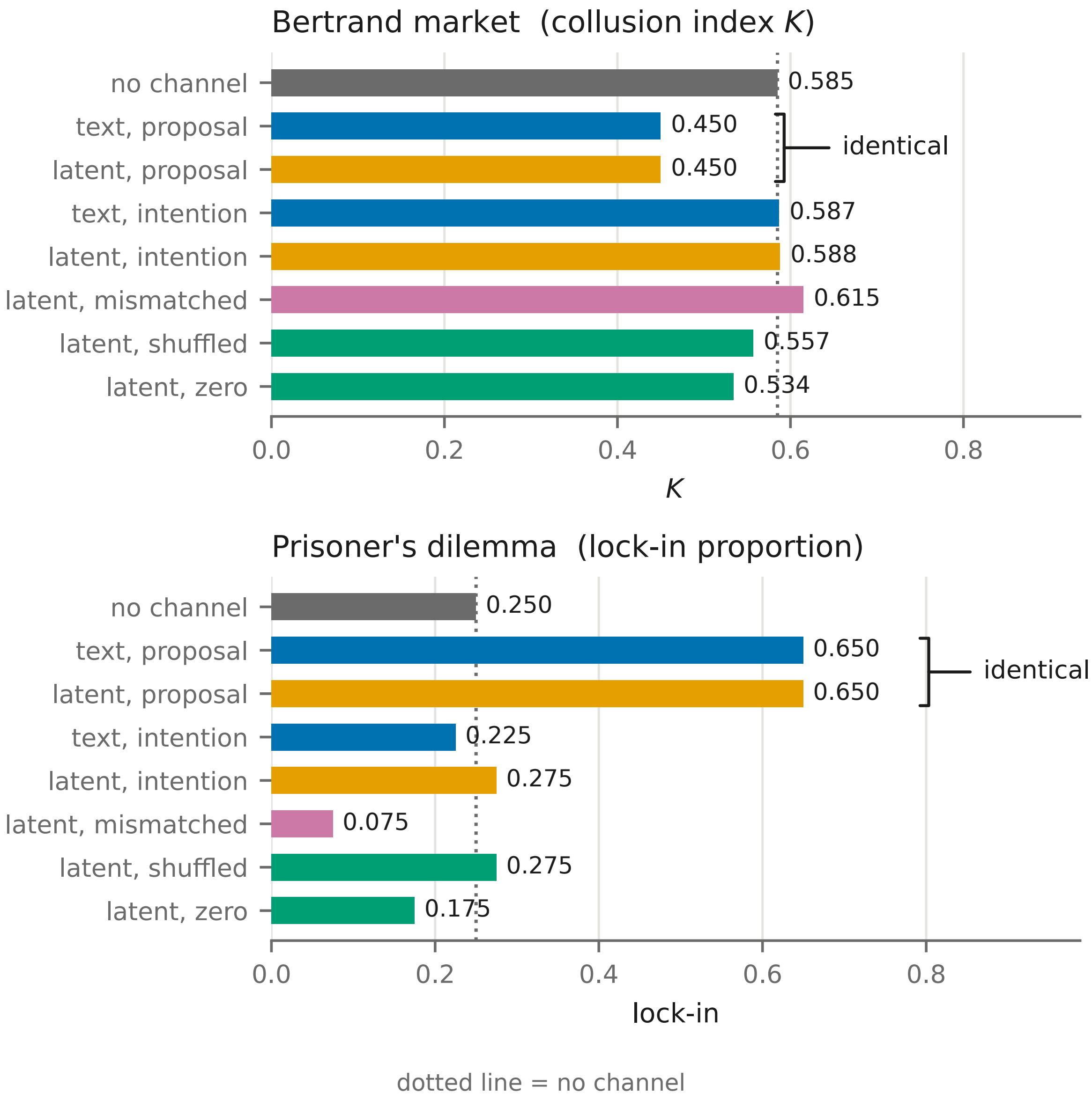

Figure 5: The effect crosses the unreadable channel. Text (blue) and latent (orange) proposal arms coincide to three decimals in both games. Inert controls (green) sit near no-channel; the content-mismatched payload (pink) breaks the effect and, in the dilemma, falls below silence—a proposal naming the wrong option induces miscoordination rather than none. Arms are identified by position and label, not by colour alone.

Table 12: The market effect across three model families, 20 seeds per arm, matched seeds and prompts. Bracketed values are bootstrap 95% intervals. Pooling is random-effects (DerSimonian–Laird): between-model variation is the quantity of interest here, and a fixed-effects pool would treat it as noise.

model	proposal − none	proposal − intention
Qwen2.5-14B	−0.126 [−0.203, −0.054]	−0.111 [−0.181, −0.047]
Llama-3.1-8B	−0.041 [−0.061, −0.021]	−0.031 [−0.054, −0.006]
OLMo-2-13B	−0.075 [−0.125, −0.028]	−0.058 [−0.119, −0.003]
pooled	**−0.069** [−0.113, −0.026]	**−0.058** [−0.103, −0.014]
no-channel baseline K: Qwen 0.580, Llama 0.827, OLMo 0.650		
heterogeneity $I^2 = 65\%$ / 61%; all three same sign		

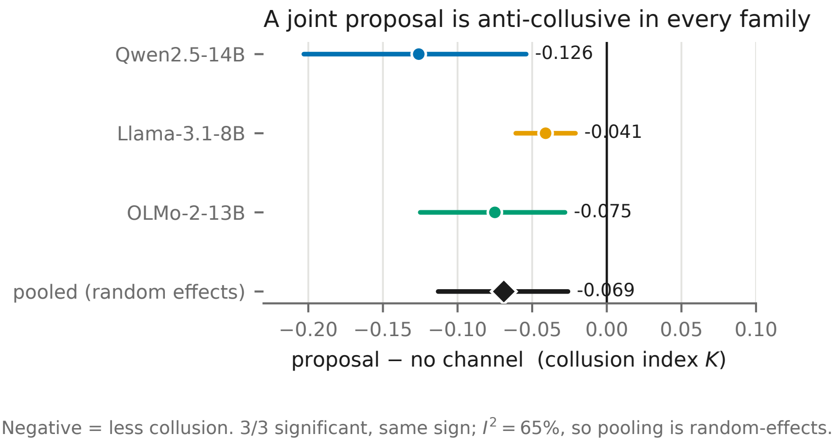

Figure 6: The market contrast on three independent model families. Points are per-model estimates with bootstrap 95% intervals; the diamond is the random-effects pool. All three fall left of zero. Heterogeneity is substantial ($I^2 = 65\%$) and is reported rather than pooled away.

Table 13: V1: detector vs. judge, n=300 stratified. Rates are conditional on the stratification, not population rates; the population base rate of detector-positive messages is 0.006.

	judge: rule	judge: not rule
detector: rule	127	23
detector: not rule	1	149